T12

$H_0: p_1 = p_2 = \frac{1}{4}, p_3 = \frac{1}{6}, p_4 = \frac{1}{3}$

$H_1: p_i = \frac{1}{4}, i = 1, \dots, 4$

$\alpha = 0,2$

H_0:

p	1	2	3	4
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{24}$	$\frac{1}{12}$
2	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{24}$	$\frac{1}{12}$
3	$\frac{1}{24}$	$\frac{1}{24}$	$\frac{1}{36}$	$\frac{1}{18}$
4	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{18}$	$\frac{1}{9}$

H1:

P	1	2	3	4
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
2	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
3	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
4	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$

(x_1, x_2) - выборка с результатами

$$l(x) = \frac{P(x|H_1)}{P(x|H_0)}$$

$G: \{l \geq c\}$

$l = \frac{1/16}{1/36} = \frac{36}{16}$ - исход $(3,3)$

$P(l \geq \frac{18}{8} | H_0) = \frac{1}{36} < \alpha$

$l = \frac{1/16}{1/24} = \frac{24}{16} = \frac{3}{2}$ - исходы $(1,3)$ $(3,1)$ $(3,2)$ $(2,3)$

~~$l = 1/16$~~

$P(l \geq \frac{3}{2} | H_0) = \frac{7}{36}$

$P(\ell \geq \frac{3}{2} | H_0) < \alpha$

$\ell = \frac{1116}{1118} = \frac{9}{8}$ - исходот $(3,4)$ и $(4,3)$

$P(\ell \geq \frac{9}{8} | H_0) = \frac{11}{36}$

$P(\ell \geq \frac{9}{8} | H_0) > \alpha$

$\alpha_2 = 1 - 5 \cdot \frac{1}{16} = \frac{11}{16}$

$P(\ell \geq \frac{3}{2} | H_1) = \frac{5}{16}$

$\{(1,3), (3,2), (3,1), (3,3)\}$

крит. обл